Eutectic-Free MicroLED and OTFT Integration: A Scalable Solution Across Display Generations

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Abstract

We present a scalable, eutectic-free integration strategy for flexible active-matrix micro-LED displays driven by organic thin-film transistors (OTFTs). Using a chip-first approach, micro-LEDs were directly assembled onto a transparent, flexible 25 µm polyethylene naphthalate (PEN) substrate, followed by low-temperature (<120 °C) OTFT backplane fabrication. This monolithic process eliminates the need for eutectic bonding or die-level repair typically required in chip-last methods. The resulting high-resolution displays exhibit stable video-rate performance before and after release from a rigid carrier. A high-yield via contact process enables robust frontplane–backplane electrical interconnection, supporting seamless integration across display sizes and form factors. This work demonstrates a versatile platform for next-generation displays, unifying flexibility, transparency, and manufacturability without reliance on conventional bonding schemes.

1. Introduction

Micro-LED technology can enable bright, efficient, high contrast displays with long lifetime and, as such, has been the subject of intense development work in both industry and academia (1-3). Initial uses of micro-LED have been in high-end TV displays (4) and augmented reality (AR) glasses (5). Other proposed uses are in transparent displays (6) and smartwatches (7). Active-matrix display driving has advantages over simpler passive-matrix drive schemes in that displays can have more rows and columns, can be brighter with higher contrast, and offer significant power savings (8). Active-matrix driving requires at least 2 transistors and 1 capacitor within the sub-pixel, resulting in a multi-mask patterning process. Due to the current driving requirement of emissive displays such as micro-LED, the light output of the display is proportional to the current flowing in the drive transistors. Stress effects due to long periods of constant bias on the drive transistor can cause changes in luminance, which can exhibit as a memory effect, causing “mura” and “ghosting” in displayed images (9). For this reason, low-temperature polysilicon (LTPS) is commonly chosen as a thin-film transistor (TFT) technology for driving micro-LED due to its high charge mobility and good bias stress stability (10). LTPS is processed at a low temperature in comparison to crystalline silicon, but the process temperature for the silicon nitride insulator and post-annealing used is around 450°C (11). Consequently, the process allows for rigid transparent displays (12-14) but does not support flexible transparent formats (15).

In addition to the high-temperature constraint of current micro-LED display systems, the process of manufacturing has also to mature to meet manufacturing readiness. Most displays are made by attaching the individual micro-LED dies to each display sub-pixel and forming a eutectic bond through thermal or laser bonding (16-18). This typically involves 3 transfer steps and a laser sintering process. Low-melting-point metals should be deposited on the anode and cathode contact pads of the backplane so that the joining process can proceed. Some of these processes are not scalable in display backplane manufacturing lines, and yield rates are still too low for products to be made in large volumes. Re-work of faulty or misplaced dies is necessary to repair the display, with hundreds or even thousands of randomly located pixels re-worked in order to meet specifications. The time required for re-work adds a high cost to the display manufacturing process, and consequently, the market for micro-LED is small due to the high costs of the initial products.

2. uLED chip last process and bonding yield

The chip-last uLED integration process consists of three main stages: laser transfer, laser bonding, and laser-based repair. Figure 1 illustrates the laser transfer process used to assemble the micro-LED array. In this step, blue, green, and red uLED dies were selectively transferred by laser from a dense donor carrier (COC1) onto a secondary carrier (COC2). While the uLED dies on COC1 were arranged in a high-density array, the dies on COC2 were redistributed to match the target pixel pitch of the final display. Due to non-ideal die yield on COC1, incomplete transfer and particle-related defects resulted in a typical COC2 yield of approximately 99.5% (17). Particles, empty dies are the possible reasons and impact the yield rate. The uLED array on COC2 was integrated onto an LTPS TFT backplane using a laser bonding process. The COC2 and LTPS substrates were precisely aligned and locally heated by laser irradiation, enabling metal interdiffusion and eutectic formation at the bonding interface. This laser-assisted bonding process achieved a bonding yield exceeding 99.99%, demonstrating robust electrical and mechanical interconnection between the uLEDs and the LTPS backplane.

To further improve display yield, a laser-based mass repair process was employed. After automatic optical inspection (AOI) identified missing or defective pixels, individual uLED dies were selectively transferred and bonded to the designated coordinates using the same laser transfer and laser bonding techniques. Although the repair step is inherently slower due to die-by-die manipulation, a repair yield exceeding 99% was achieved. As a result, the overall display yield after repair reached approximately 99.999%. For example, in a smartwatch display comprising approximately 400,000 sub-pixels, an initial yield of 99.5% corresponds to about 2,000 defective dies requiring replacement.

Organic thin-film transistors (OTFT) have been integrated with micro-LED in a chip-first approach through monolithic processing on top of a sapphire wafer, which had micro-LEDs processed upon it (18). This demonstrated the possibility of reversing the order of the processing of TFTs and micro-LEDs when a low-temperature backplane was used (150°C maximum temperature). Planarisation on top of flip-chip micro-LEDs, followed by photolithographic patterning of vias through the planariser, coupled with metal

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sputtering of conductors down the via, again patterned by photolithography, was used to make an electrical connection between the micro-LEDs and the TFT backplane. Via processes like the ones used in that approach are commonplace in microelectronics and have been used for decades and with extremely high yield (fault rate likely less than 1 in 10^9 connections), hence re-using the technique to make electrical connections to millions of micro-LEDs on a display substrate is an attractive proposition for solving the low yield of connections between TFT and micro-LED described previously. Inorganic TFT technologies (LTPS and oxide TFT), which use the high-temperature silicon nitride PECVD process, cannot easily be used to make displays since the high-temperature process would affect the micro-LED devices (which are themselves temperature sensitive and cannot withstand processing above 300°C without serious degradation.

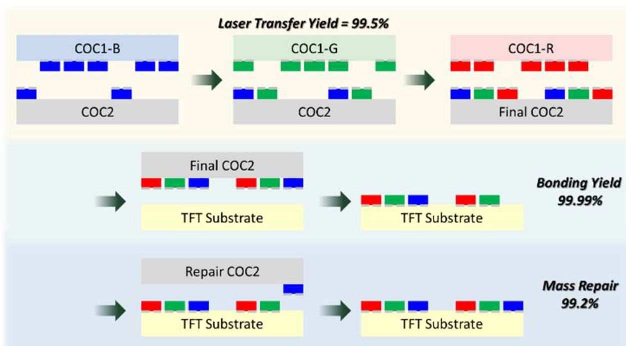

Figure 1. Schematic of the chip's last process of uLED transfer (top, transfer, middle bonding, bottom repair).

3. Chip-first display fabrication

This paper presents results of a chip-first micro-LED active matrix OTFT display on an optically clear plastic substrate. Micro-LED devices were processed upon a sapphire substrate according to the scheme shown in Figure 2.

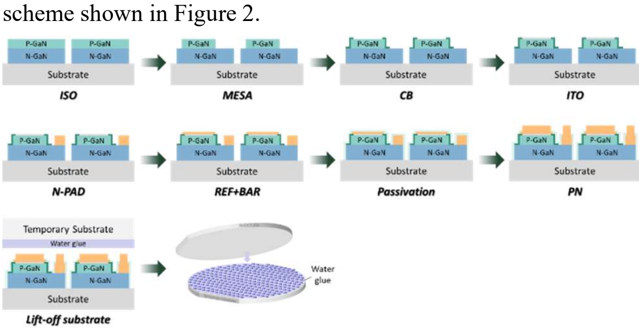

Figure 2. Micro-LED fabrication process.

The chemical lift-off technology of GaN epitaxial layers from sapphire substrates is an innovative semiconductor manufacturing process aimed at addressing the heat dissipation issues present in traditional sapphire substrate technologies used in high-end optoelectronic devices, electronic power devices, and microwave radio frequency devices.

Firstly, a dielectric patterned sapphire substrate (referred to as DPSS substrate) is fabricated on a planar sapphire substrate. Then, using an MOCVD device, high-quality GaN epitaxial layers are grown on the DPSS substrate through epitaxial lateral overgrowth techniques. The epitaxial layer structure may include multiple key components such as P-type GaN, quantum wells, and N-type GaN.

Next, photolithography, plasma etching, and vapor deposition technologies are used to finely process the GaN epitaxial structure, ultimately fabricating it into a micro-LED chip. After the chip production is completed, chemical lift-off technology is further utilized to achieve precise separation of the GaN chip from the sapphire substrate. It is particularly noteworthy that our chemical lift-off technology ensures perfect separation of the GaN chip from the sapphire substrate and also guarantees that there will be no chip peeling-off phenomenon.

Finally, organic glue with strong adhesion is used to transfer the micro-LED chip from the sapphire substrate to a temporary substrate. Thus, the separation process of the GaN chip from the sapphire substrate is successfully completed.

Transfer of the micro-LED to a plastic substrate and fabrication of the active-matrix OTFT backplane are shown in Figure 3 (a), starting with the coating of a UV-acrylate pre-polymer layer, acting as a glue. The micro-LED substrate is pressed into the liquid pre-polymer followed by UV curing to adhere the micro-LED layer to the plastic substrate. Removal of the temporary substrate is achieved by dissolving the glue layer in water or acetone (2 types of organic glue are available). Some mechanical peeling is required in the case of the water glue to help achieve full separation. An additional layer of acrylate planarization is then UV-cured and patterned using positive photoresist as a mask, followed by dry etching to form via holes in register with the contact pads of the micro-LEDs. Metallization using sputter coating and patterning by photolithography then makes a connection from the micro-LED to the layer on the planarization. The LEDs can be tested at this point to verify connections are successfully made.

The process continues with the process as described in reference (18) to build up the OTFT backplane. As a final stage the glass carrier can be removed from the thin (25 um) PEN substrate to yield a flexible transparent micro-LED display.

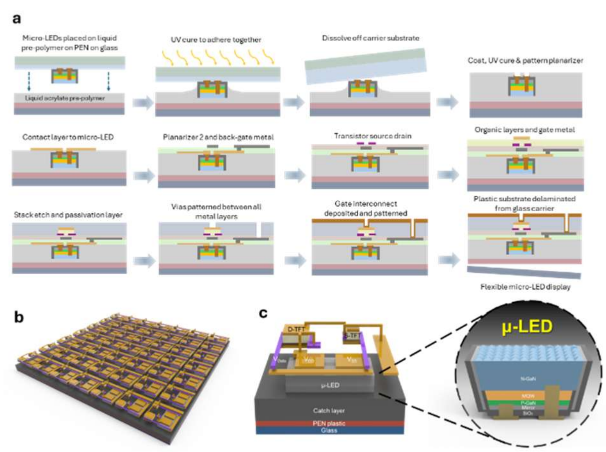

Figure 3. (a) Transfer of micro-LED to plastic substrate, planarization, contact metallization, and fabrication of OTFT backplane - the" chip-first" route. (b) Array of OTFT devices (c) Cross-section of a single pixel.

4. Low-temperature OTFT Circuit Design and TFT Performance Characterization

Figure 4 presents the design and electrical characterization of the active-matrix TFT pixel circuit used for driving micro-LEDs. Panel (a) illustrates the pixel circuit schematic, which includes a select TFT, a drive TFT, and a storage capacitor (C_{st}). The select TFT samples the data voltage during the scan period and stores it on the capacitor, which subsequently biases the gate of the drive TFT to control current flow to the micro-LED. The corresponding layout design is shown in panel (b), and the fabricated structure is presented in the optical micrograph in panel (c), highlighting the pixel integration and interconnect routing.

Electrical characterization results are shown in panels (d)-(f). The transfer characteristics in panel (d) demonstrate low off-state leakage (<10^{-12} A) and a sharp subthreshold slope across various drain voltages ($V_D = -1$ V to -13 V). Panel (e) shows the extracted field-effect mobility, which reaches a peak of approximately 2.1 $cm^2/V{\cdot}s$ at $V_D = -1$ V. This mobility value is sufficient for driving micro-LEDs with the required current levels, particularly in high-resolution, low-power display configurations.

The output characteristics in panel (f) confirm the functional behavior of the drive TFT under different gate voltages ($V_G = 3$ V, 6 V, 9 V), showing current saturation and proper modulation of drain current. The drive current levels achieved are adequate for micro-LED pixel emission, verifying that the designed circuit meets the operational requirements for active-matrix display driving. The results collectively demonstrate that the TFT performance and circuit design are well-suited for integration in micro-LED display systems.

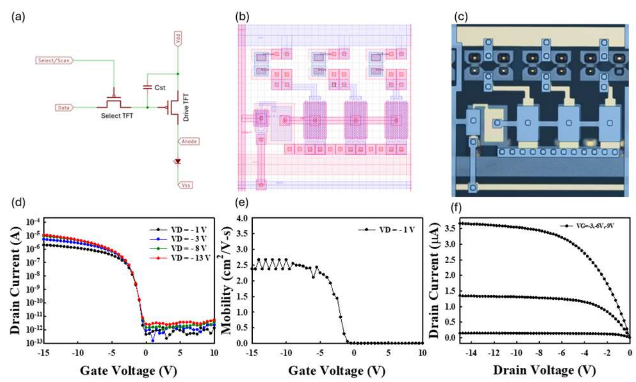

Figure 4. Transfer curves for the drive and select TFTs for the micro-LED 2T-1C pixel circuit.

5. High-Yield Chip-first micro-LED Architecture Enabled by Via-Based Interconnects

Figure 5 illustrates the proposed single-sided micro-LED display architecture, which addresses several critical challenges in conventional micro-LED display manufacturing. Image (a) shows an array of micro-LEDs that are first placed directly onto the substrate, prior to via formation. This uLED-first approach eliminates the need for mass transfer processes, significantly simplifying fabrication and improving alignment accuracy. Unlike traditional methods that rely on pick-and-place or laser-induced transfer steps with low yield, this approach avoids any metallurgical bonding, thereby reducing both mechanical stress and alignment error. As revealed in the SEM image (b), the device employs a planar and highly ordered via configuration that allows for precise signal routing without wrap-around wiring, simplifying the fabrication process and reducing interconnect complexity.

Panel (c) shows a fully assembled and functional large-area display under operation, emitting uniform blue light across the active area. Notably, the image confirms that no visible tile joins or mismatched seams are present, which typically plague tiled displays due to panel-to-panel alignment issues. This seamless output is enabled by the monolithic single-panel design, allowing the entire display to be fabricated in a single process without the need for post-assembly calibration or visual matching. This characteristic not only improves visual uniformity but also greatly enhances manufacturability for ultra-large displays.

The cross-sectional SEM in panel (d) highlights the vertical via structure used for electrical connection, demonstrating precise alignment and robust mechanical integration between the LED chip and substrate. A total of 18,000 micro-LEDs were successfully interconnected using this via architecture, with nearly zero via connection failures observed during testing—highlighting the high reliability and yield of the process. This via-based approach supports a manufacturing yield exceeding 99.99%, enabling a “First Time Right” process with minimal or no rework. Furthermore, the single-sided configuration results in a lightweight and ultra-thin display module that can be directly laminated onto a metal heatsink layer for effective thermal dissipation—especially critical in high-brightness or large-area wall-mounted applications. Finally, due to the absence of aging-prone encapsulation materials, this design is highly resistant to yellowing, making it ideal for transparent or flexible display formats. Altogether, the proposed architecture represents a scalable and manufacturable path forward for next-generation micro-LED displays.

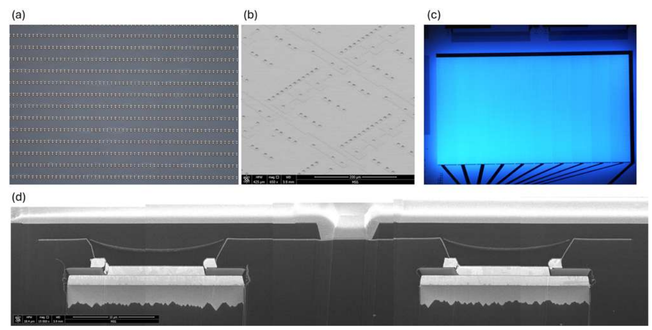

Figure 5. Single-sided micro-LED architecture: (a) uLED-first array on substrate. (b) Planar via layout (SEM). (c) Uniform large-area blue emission. (d) Cross-sectional SEM of vertical via interconnections.

6. Display bonding and driving

The displays were electrically interfaced using anisotropic conductive film (ACF) and flexible printed connectors (FPC), which connected the panel to a 48×48 row/column driver matrix for video display. Display driving was implemented using an external FPGA-based control board, which generated the necessary row and column scanning signals as well as data input waveforms. Gamma correction and timing control were pre-configured on the FPGA, enabling accurate gray-scale image rendering on the micro-LED display.

To achieve mechanical flexibility, the PEN substrate was delaminated from the carrier glass through manual peeling at 120 °C. The resulting semi-transparent, flexible display is shown in operation and being flexed in Figure 6, demonstrating the device’s mechanical compliance and functional integrity after substrate release.

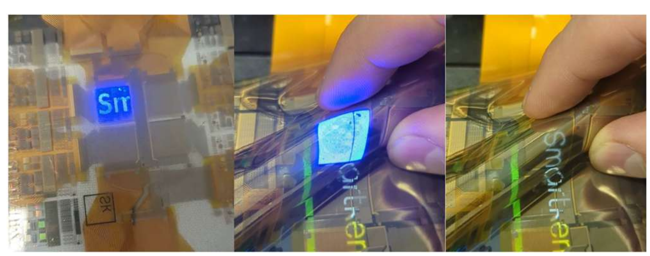

Figure 6. Image of driving micro-LED chip-first display.

7. Conclusions

This work presents a scalable, eutectic-free integration strategy for active-matrix micro-LED displays using a chip-first approach combined with low-temperature organic thin-film transistor (OTFT) technology. By fabricating the OTFT backplane below 120 °C, the process becomes fully compatible with flexible and transparent polymer substrates, and crucially, enables monolithic integration of micro-LEDs without thermal damage—something not feasible with conventional high-temperature TFT technologies such as LTPS or oxide-based systems.

The chip-first architecture eliminates the need for mass transfer and eutectic bonding, addressing major bottlenecks in yield, cost, and scalability. A high-yield via interconnection process (>99.99%) successfully connected over 18,000 micro-LEDs without failure, confirming the robustness of the frontplane–backplane integration. The 2T-1C OTFT pixel circuit exhibited sufficient mobility (~2.1 $cm^2/V\cdot s$) and output current for micro-LED driving, while FPGA-based external control confirmed full gray-scale and gamma-corrected display operation.

Altogether, the demonstrated platform offers a reliable and manufacturable solution for next-generation micro-LED displays, enabling seamless integration on thin, transparent substrates without relying on high-temperature or rework-intensive processes.

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